

Determination of Rare Earth Elements in Geological Samples Using Laser-Induced Breakdown Spectroscopy (LIBS)

Applied Spectroscopy
2018, Vol. 72(1) 114–121
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DOI: 10.1177/0003702817734854
journals.sagepub.com/home/asp


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Abstract

Laser-induced breakdown spectroscopy (LIBS) was used to detect rare earth elements (REEs) in natural geological samples. Low and high intensity emission lines of Ce, La, Nd, Y, Pr, Sm, Eu, Gd, and Dy were identified in the spectra recorded from the samples to claim the presence of these REEs. Multivariate analysis was executed by developing partial least squares regression (PLS-R) models for the quantification of Ce, La, and Nd. Analysis of unknown samples indicated that the prediction results of these samples were found comparable to those obtained by inductively coupled plasma mass spectrometry analysis. Data support that LIBS has potential to quantify REEs in geological minerals/ores.

Keywords

Rare earth elements, REE, geological samples, laser-induced breakdown spectroscopy, LIBS

Date received: 23 May 2017; accepted: 4 September 2017

Introduction

The group of 15 lanthanides with atomic numbers from 57 to 71 (La to Lu) and two other elements scandium (Sc) and yttrium (Y) with atomic numbers 21 and 39, respectively, are categorized as rare earth elements (REEs). Because of their unique physicochemical properties, REEs have become important for both scientific and technological point of views. Scientifically, the elements have been widely used in geochemistry because the knowledge of their distribution in rocks and minerals, and their mobilization under different environmental conditions can be used to sort out important geochemical and petrogenetic processes.¹ The economic interest of REEs is evident from the fact that the elements play an important role in various fields and considered essential elements in many industrial applications.² The REEs are currently in high demand due to their application in defense and importance in renewable energy industries as well as their use in high-tech products such as cell phones, magnets, fiber optic cables, television, hybrid cars, and medical imaging.

The most commonly used techniques in the determination of REEs include inductively coupled plasma mass spectrometry (ICP-MS), inductively coupled plasma optical emission spectrometry (ICP-OES), neutron activation analysis (NAA), and X-ray fluorescence (XRF).³ Inductively coupled plasma mass spectrometry and ICP-OES require

time-consuming sample preparation–dissolution using either corrosive acids or fusion with fluxes. For direct analysis of solid samples, these techniques need to be coupled with a laser ablation sampling technique.⁴ Neutron activation analysis is a very sensitive technique, but it requires access to a nuclear reactor and a long irradiation time for the detection of REEs, especially direct analysis of solid samples. X-ray fluorescence also offers the possibility of direct analysis of solid samples. However, the technique is not suitable for lighter elements and wavelength dispersive XRF (WDXRF) spectrometers are used to determine REE rather than energy dispersive XRF (EDXRF) instruments. For rapid detection and analysis of REEs, laser-induced breakdown spectroscopy (LIBS) can be used. Laser-induced breakdown spectroscopy is an in situ spectroscopic

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analytical technique that requires little to no sample preparation. High-power laser energy is focused on the sample to be analyzed; spectral signatures of characteristic elements present in the sample are collected and analyzed to extract the necessary information. Laser-induced breakdown spectroscopy has been widely used for the detection and quantification of elements in materials irrespective of their states.^{5–12} There are a limited number of reports in the literature showing the application of LIBS for the study of REEs. Most of the published studies are either on the standard samples or samples mixed with pure oxides of REEs.^{13–18} Abedin et al.¹⁹ and Phuoc et al.²⁰ have performed qualitative analysis and identified some REE peaks in monazite sand and coal ash, respectively. Bhatt et al.¹⁸ performed qualitative and quantitative analysis of six REEs (Ce, Eu, Gd, Nd, Sm, Y) using their pure oxides as working samples.

In this study, we have used natural REE ores/minerals for the detection and quantification of these elements. To our knowledge, this appears to be the first study to demonstrate the application of LIBS in the quantification of REEs in geological samples. Eight samples with varying concentrations of REEs were studied. Several emission lines for each REE were detected in the LIBS spectra obtained from these samples. For quantification, multivariate analysis (MVA) was performed. Partial least squares regression (PLSR) calibration models were developed by using seven samples as training while one was used as unknown to test the validity of developed calibration models.

Experimental Setup

A schematic of the experimental set-up is shown in Fig. 1. Laser pulses from a Q-switched neodymium-doped yttrium

aluminum garnet (Nd:YAG) laser (Tempest, New Wave Research) operating at 1064 nm wavelength, 5 ns pulse duration, and 1 Hz repetition rate were focused on sample pellets placed on a uniformly rotating platform. The platform was rotated so that each laser shot would interact with fresh sample. The beam diameter of laser output was 5 mm. Argon gas was continuously supplied to the sample surface uniformly so that ablated aerosols will be swept out and created plasma would be produced in Ar atmosphere. Moreover, to provide a homogeneous Ar atmosphere around the plasma, this whole sample platform was enclosed by a small enclosure. The laser output energy was varied by means of a pair of half-wave plate and polarizing beam splitter cube (60:40). The pulse energy was monitored throughout the measurements using an Ophir pyroelectric high-energy sensor (PE25BF-DIF-C) and a compact Juno USB interface connected to the computer. An N-BK7 plano-convex lens of 10 cm focal length was used to focus the laser beam onto the sample surface at normal incidence. Plasma emission was collected through a telescope made of two ultraviolet (UV)-grade fused silica plano-convex lenses of 5 cm and 1 cm focal length, and an ultraviolet visible (UV-Vis) fiber optic of 1000 μm core diameter and 2 m long (Ocean Optics). The optical fiber was connected to an Echelle spectrograph (Aryelle 200, LTB Lasertechnik Berlin), which was coupled with a 1024 $\times$ 1024 intensified charge coupled device (ICCD) camera (PI-MAX4, Princeton Instruments).

Sample Preparation

Eight geological samples (S1–S8) chosen from our laboratory collection were dried and powdered in mortar and

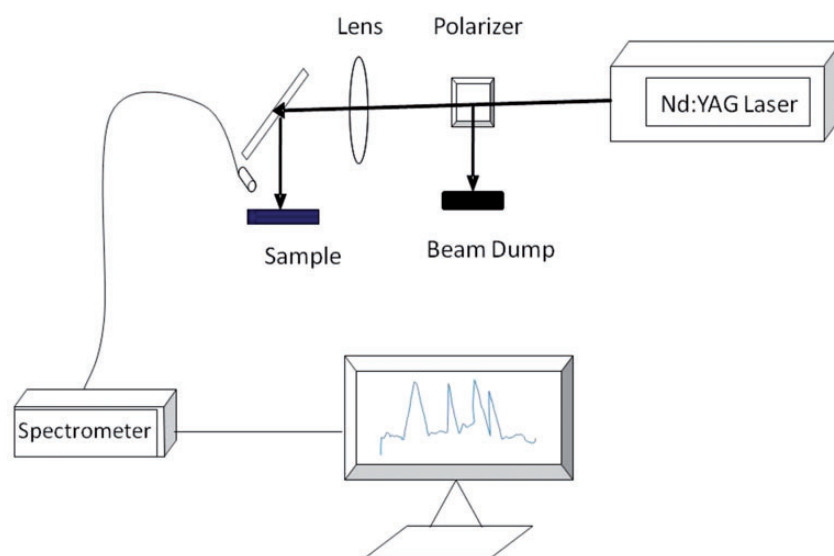


Figure 1. Laser-induced breakdown spectroscopy experimental set-up for geological samples.

Table 1. Rare earth element composition of geological samples as determined using ICP-MS.

Samples	Concentrations (PPM)															
	Sc	Y	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
S1	39	100	814	1367	118	429	55	14	51	6	28	4	12	1.4	9	1.3
S2	43	136	1103	1880	165	590	78	19	72	8	37	5	16	1.8	12	1.6
S3	71	293	1366	2193	212	740	89	22	93	11	63	11	32	3.9	24	3.2
S4	45	169	1479	2284	221	786	102	25	92	10	48	7	20	2.2	14	1.9
S5	45	196	1779	2685	265	957	126	31	113	12	59	8	23	2.5	16	2.1
S6	51	202	1794	2751	265	964	126	31	113	12	58	8	23	2.7	16	2.3
S7	44	206	1922	2861	285	1027	136	33	119	12	60	9	24	2.7	17	2.3
S8	14	750	26556	57513	7710	37833	4835	633	2552	201	711	36	199	5.1	32	3.5

Table 2. Concentration of Ce, La, and Nd in pellets (including 10% starch as binder) prepared for quantitative analysis.

Samples	Concentration (%)		
	Ce	La	Nd
S1-M	0.89	0.85	0.81
S2-M	2.48	2.42	2.37
S3-M	4.05	4.00	3.93
S4-M	7.90	7.88	7.80
S5-M	10.79	10.79	10.70
S6-M	17.84	17.89	17.78
S7-M	21.39	21.46	14.22
S8	5.22	2.41	3.43

pestle. Part of each sample was used for ICP-MS measurement; the data are presented in Table 1. For LIBS analysis, pellets of 13 mm diameter were prepared using 0.5 g of powdered sample and 0.05 g of starch as binder using bench-top manual pellet press (Carver). A pressure of 6.5 tons was applied to prepare the pellets.

For quantitative analysis, different amount of pure oxides of Ce, La, and Nd were added to the seven samples (S1–S7) to make a wider range of concentration and the modified samples with added REEs are denoted by S1-M, S2-M, S3-M, S4-M, S5-M, S6-M, and S7-M (Table 2). The final concentration of Ce, La, and Nd in these samples varied in the ranges of 0.89–21.39%, 0.85–21.46%, and 0.81–17.78%, respectively (Table 2). No addition of REE oxides was made to the sample S8.

Results and Discussion

Qualitative Study

Inductively coupled plasma analysis showed that all the samples except S8 had less than 0.3% (3000 ppm) concentration

of each REE, S8 had larger concentration of Y, La, Ce, Pr, Nd, Sm, Eu, Gd, and Dy in the range of 0.06–5.7% (633–57513 ppm). Because of the low concentrations, spectral lines of REEs tended to have a lower intensity. Hence, spectra were recorded in Ar atmosphere to see if intensity of the spectral lines could be enhanced. As expected, spectral peaks were found to be more intense in Ar atmosphere than in ambient air as shown in Fig. 2a. Since the natural geological samples have complex matrix, the spectra obtained from these samples contained a large number of crowded and overlapping spectral peaks as shown in Fig. 2b. We also recorded spectra from pure REE oxides to locate the wavelength position of different emission lines of these elements.

For the identification of spectral lines, we considered peaks with intensity levels higher than three times the noise (i.e., 3σ using International Union of Pure and Applied Chemistry criterion). To confirm that a peak corresponds to a particular emission line, the wavelength positions found at the NIST database, including those observed in spectra obtained from pure samples, were taken as reference. Moreover, a peak was only considered if it was observed in all the ten spectra obtained from the sample with a consistent peak structure. To minimize shot-to-shot fluctuation, 50 acquisitions were averaged to get a single spectrum. We checked for persistent emission lines of REEs given by the NIST database within the wavelength range of 200–800 nm to identify these elements. We identified most of the persistent lines of Ce, Nd, and La (presented in Table 3) in the spectra obtained from all the samples. In the spectra recorded from the sample S8, some emission lines of Y, Pr, Sm, Eu, Gd, and Dy (presented in Table 4) were also detected. Some of the emission lines observed in the spectra recorded at 2 μ s gate delay and 7 μ s gate width with 45 mJ laser energy are shown in Figs. 3 and 4. Many REE emission lines were observed to suffer interferences from the major elements present in geological samples. We were not able to detect any emission lines of the elements Sc, Tb, Ho, Er, Tm, Yb, and Lu in the spectra obtained from

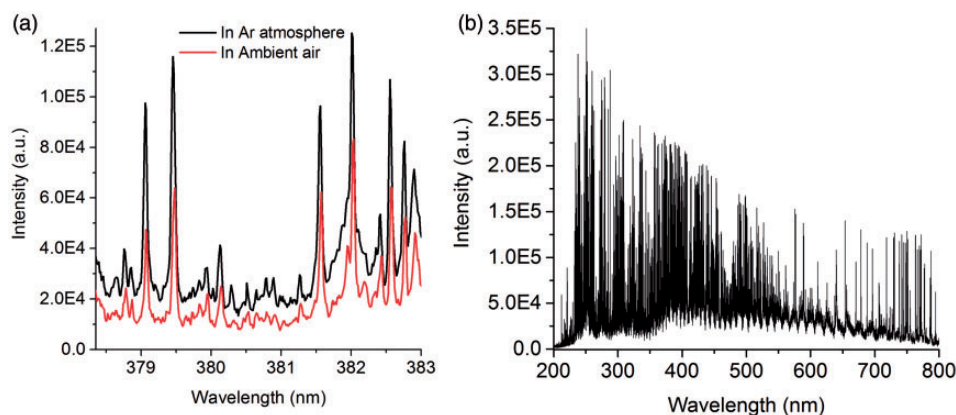


Figure 2. (a) Spectra (in the range of 377–383 nm) recorded in Ar atmosphere and ambient air. (b) Full range (200–800 nm) spectra recorded from sample S1.

the given samples, probably because of their low concentrations and poor resolution of the spectrometer.

Table 3. Identified emission lines of Ce, Nd, and La and corresponding samples in which they were detected.

Identified lines	Samples in which these lines were detected
Ce(I) 560.12	S1, S2, S3, S4, S5, S6, S7, S8
Ce(I) 571.90	S2, S3, S4, S5, S6, S7, S8
Ce(I) 594.08	S1, S2, S3, S4, S5, S6, S8
Ce(II) 380.15	S1, S2, S3, S4, S5, S6, S7, S8
Ce(II) 413.76	S1, S2, S3, S4, S5, S6, S7, S8
Ce(II) 418.65	S1, S2, S3, S4, S5, S6, S7, S8
Ce(II) 446.02	S1, S2, S3, S4, S5, S6, S7, S8
Ce(II) 456.23	S1, S2, S3, S4, S5, S6, S7, S8
Nd(II) 378.42	S3, S6, S7, S8
Nd(II) 380.53	S1, S2, S3, S4, S5, S6, S7, S8
Nd(II) 386.34	S2, S3, S4, S5, S6, S8
Nd(II) 394.15	S6, S8
Nd(II) 395.11	S1, S2, S3, S4, S5, S6, S7, S8
Nd(II) 401.22	S1, S2, S3, S4, S5, S6, S7, S8
Nd(II) 415.60	S4, S6, S8
La(I) 428.02	S1, S2, S3, S4, S5, S6, S7, S8
La(I) 514.54	S1, S2, S3, S4, S5, S6, S7, S8
La(I) 521.18	S1, S2, S3, S4, S5, S6, S8
La(I) 550.13	S1, S2, S3, S4, S5, S6, S7, S8
La(I) 624.99	S3, S4, S5, S8
La(II) 379.08	S1, S3, S4, S5, S6, S8
La(II) 399.57	S1, S2, S3, S4, S5, S6, S7, S8
La(II) 404.29	S1, S2, S3, S4, S5, S6, S7, S8
La(II) 408.67	S1, S2, S3, S4, S5, S6, S7, S8
La(II) 433.37	S1, S2, S3, S4, S5, S6, S7, S8

Quantitative Study

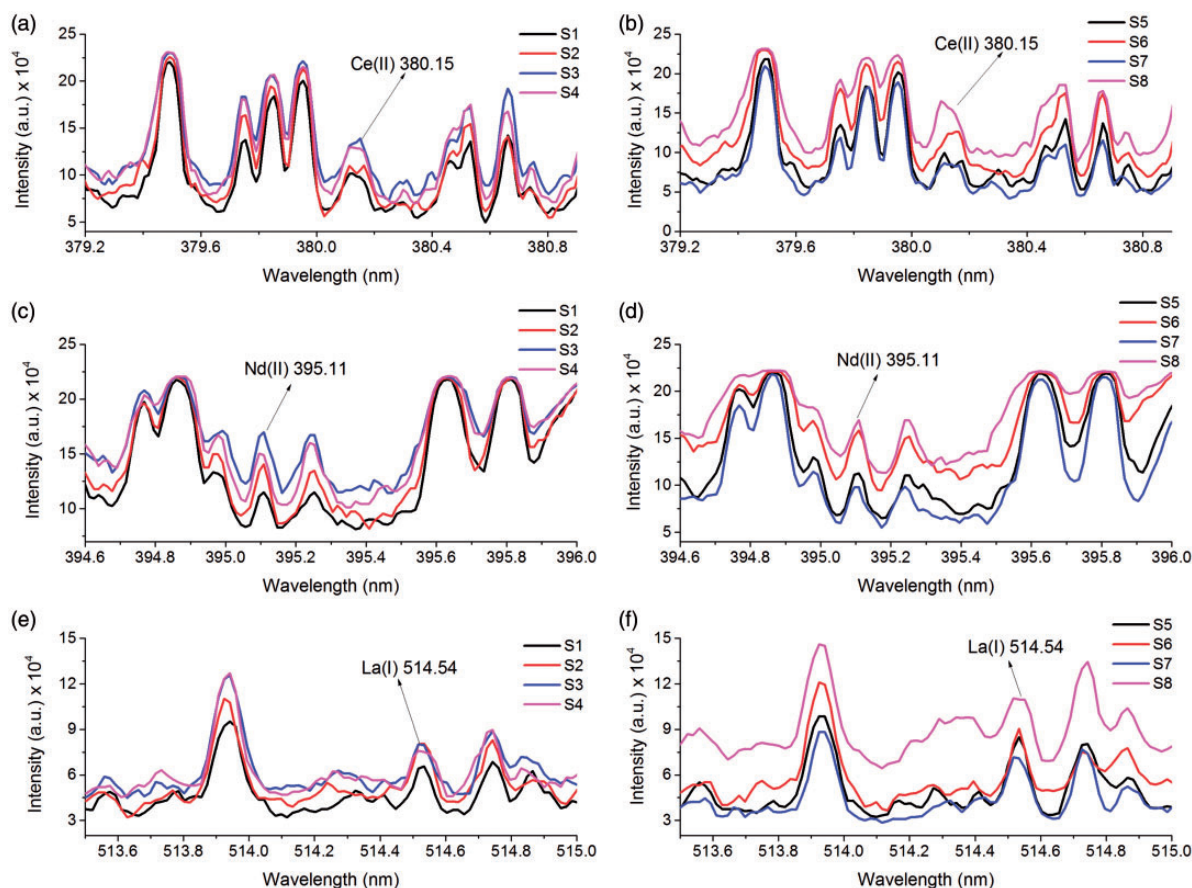
As explained in the Qualitative Study section, Ce, Nd, and La were detected in most of the samples. Therefore, these elements were selected for quantitative analysis. Except for sample S8, Ce, Nd, and La concentrations varied only with in a small range in all the samples; for example, Ce varies from 0.14% to 0.28% in seven samples (S1–S7), whereas sample S8 contains 5.75% Ce. Therefore, we could not expect significant difference in spectra from these seven samples to use them to develop calibration models. Hence, we added varying amount of pure oxides of Ce, La, and Nd to seven samples (S1–S7) so that the concentration of Ce, La, and Nd was in the range of 0.89–21.39%, 0.85–21.46%, and 0.81–17.78%, respectively, while the matrix remained the same. These new samples with added concentrations of Ce, La, and Nd were denoted by S1-M, S2-M, S3-M, S4-M, S5-M, S6-M, and S7-M. Since these samples were complex mixtures of many elements, the matrix effect could be obvious. Therefore, multivariate analysis (MVA) was executed for quantification purposes. Partial least squares regression (PLSR) models were developed by using The Unscrambler X (Version 10.3, CAMO Software). Partial least squares (PLS) is one of the multivariate analytical techniques used in chemometrics. It helps to reduce the matrix effect when dealing with complex samples.²¹ It generates a regression model that correlates the two matrices (X and Y) and finds variables in one (X) that predict variables in another (Y).²² Hence, the matrix X can be regarded as a set of predictors (n objects, m variables) and Y as a set of responses (n objects, p responses). The samples with known parameters are used to construct a model correlating two matrices **X** and **Y**, which can be later used to predict unknown parameters of other

Table 4. Spectral lines of Y, Pr, Sm, Eu, Gd, and Dy detected in the spectra obtained from sample S8.

Elements	Spectral lines
Y	324.22, 354.90, 360.07, 361.10, 363.31, 371.02, 377.43, 407.73, 412.82, 417.75, 437.49, 488.36, 490.01, 508.741 nm
Pr	417.93, 420.67, 422.29, 495.13 nm
Sm	356.82, 359.26, 363.42, 369.39, 442.43, 446.73, 447.08 nm
Eu	372.49, 381.96, 390.71, 397.19, 412.97, 420.50, 459.40, 462.72, 466.18 nm
Gd	335.04, 335.86, 342.24, 354.58, 358.49, 371.35, 374.34, 376.83, 379.63, 404.98, 405.36, 417.55, 451.96 nm
Dy	353.17, 364.53, 396.83, 404.59, 418.68 nm

samples. The appropriate number of components (factors) can be selected to obtain PLSR plots. The explained variance illustrates how much of the total variation in **X** or **Y** is described by models including different numbers of components. Ideally, models should have large (close to 100%) total explained variance with as few components as possible. The predicted versus reference plot in PLSR displays the predictive capacity of the developed models. Major statistics, which determine the quality of the regression model, are slope, offset, root mean square error (RMSE), and regression coefficient (R^2).

In this study, spectra recorded from seven samples (S1-M to S7-M) with added concentrations of Ce, La, and Nd were used to develop models and prediction capacity of those models were evaluated with the spectra obtained from the sample S8. The experimental parameters (gate delay, gate width, laser energy) were optimized by considering signal-to-noise (S/N) ratio separately for Ce, La, and Nd.

**Figure 3.** Emission lines Ce(II) 380.15, Nd(II) 395.11, and La(I) 514.54 nm observed in the spectra recorded from different samples.

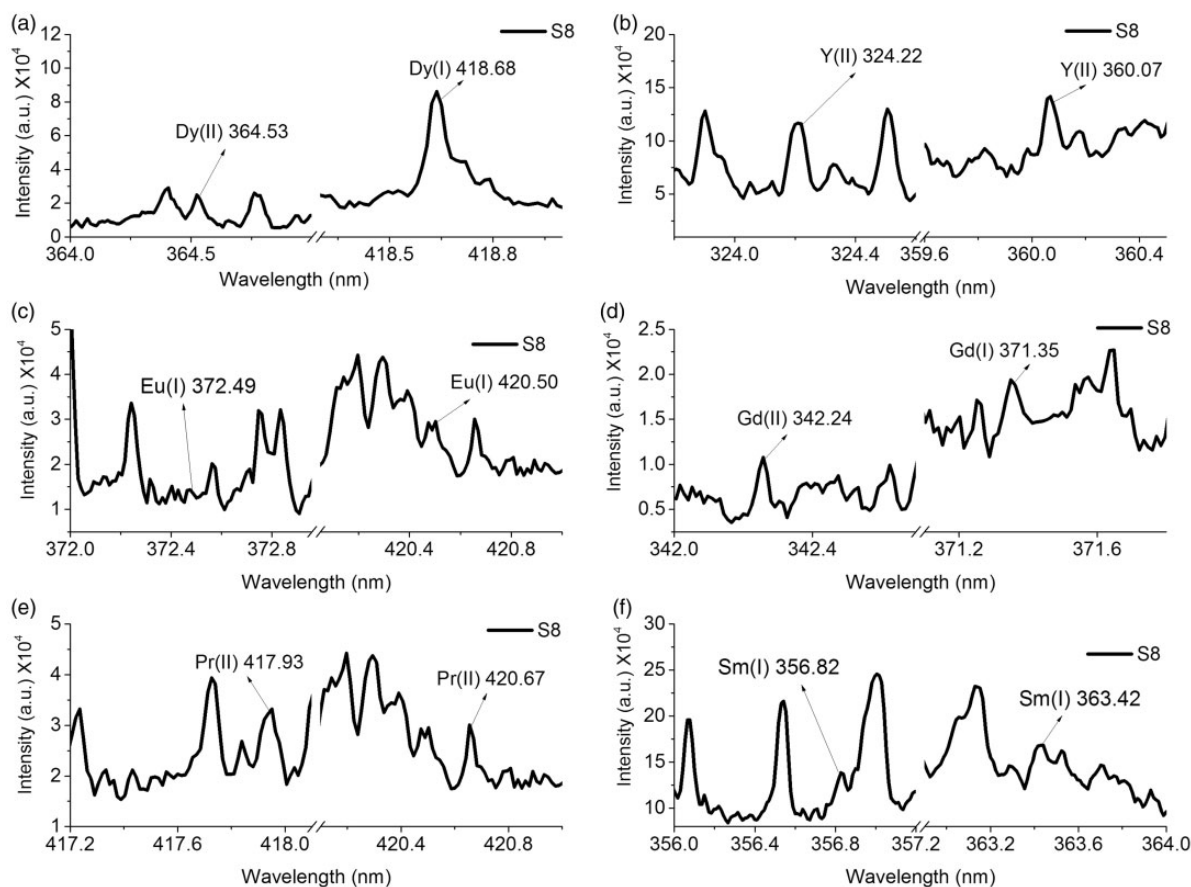


Figure 4. Emission lines Dy(II) 364.53, Dy(I) 418.68, Y(II)324.22, Y(II)360.07, Eu(I)372.49, Eu(I) 420.50, Gd(II) 342.24, Gd(I) 371.35, Pr(II) 417.93, Pr(II)420.67, Sm(I) 356.82, and Sm(I) 363.42 nm observed in the spectra obtained from the sample S8.

The optimum laser energy was 45 mJ/pulse for the three elements, while gate delay and gate width were optimized to be 2 μ s and 7 μ s for Ce, 1 μ s and 7 μ s for Nd, and 2 μ s and 4 μ s for La.

The recorded spectra in the ARY format were converted into text files to get all the wavelengths and corresponding intensities. The spectral intensities were normalized with the maximum intensity in the whole wavelength range and were taken as m variables of predictors, while concentrations were considered as p responses for PLSR. Since better results are expected by selecting a proper spectral range,²³ we used certain range of wavelengths for each element in which most of their strong emission lines were included. The selected wavelength range for Ce was 300–700 nm, La was 320–650 nm, and Nd was 345–575 nm. Partial least squares regression calibration models for the elements Ce, La, and Nd were obtained by considering seven components (factors) in the Unscrambler software and later plotted in Origin software for better visualization. Explained variances and

calibration–validation curves are shown in Fig. 5. In these figures, five spectra from each sample, which are the average of 50 acquisitions, are used to obtain plots. For all three elements, very good coefficients of determination (R^2) were obtained for both calibration and validation plots.

Prediction Capacity of Partial Least Squares Regression Models

To evaluate the prediction capacity of developed calibration models, the concentration of Ce, La, and Nd in the sample S8 were predicted by using these models. Five spectra that consisted of 50 acquisitions individually obtained from this sample were used for prediction and predicted values were compared with the values calculated from ICP-MS data. Inductively coupled plasma values were the average of five measurements while the five replicates were averaged to get LIBS values. The results are given in Table 5 and presented graphically in Fig. 6, where the error bars represent standard deviations of the measurements. The relative

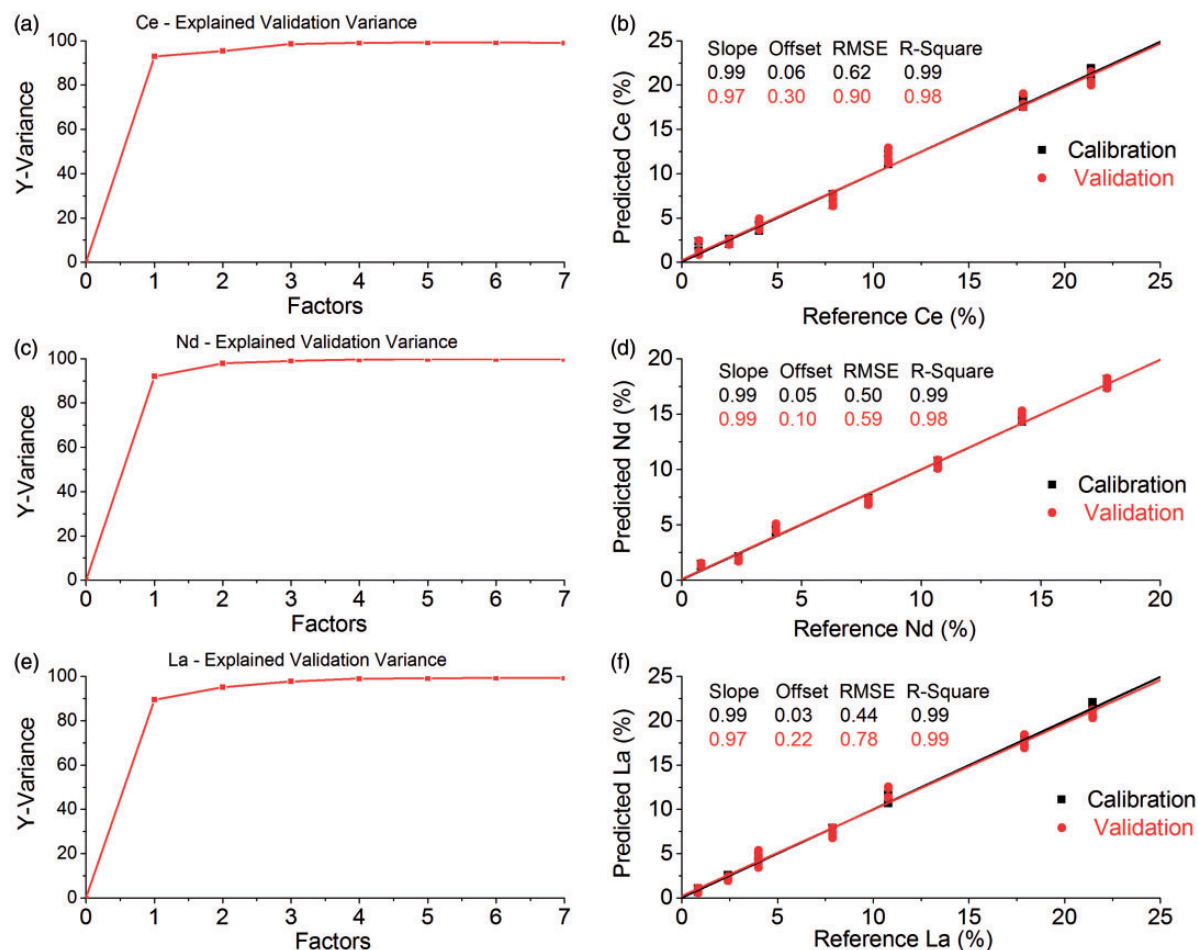


Figure 5. Explained validation variances and calibration-validation curves.

Table 5. Concentration of Ce, Nd, and La obtained using ICP-MS and LIBS.

Analyte	Concentration (%)		Relative difference (%)
	ICP-MS	LIBS	
Ce	5.22	5.58	6.76
Nd	3.43	3.19	7.09
La	2.41	2.72	12.95

differences between the ICP values and those predicted using calibration models (LIBS values) were expressed in percentage taking the ICP values as references. The values of relative differences were found to be 6.7%, 7.0%, and 12.9%, respectively, for Ce, Nd, and La. This means the concentrations of Ce, Nd, and La predicted by PLSR models are reasonably in agreement with those obtained by ICP analysis. Hence, this study demonstrated that the regression models can be produced and can be used to predict concentration of REEs in unknown samples.

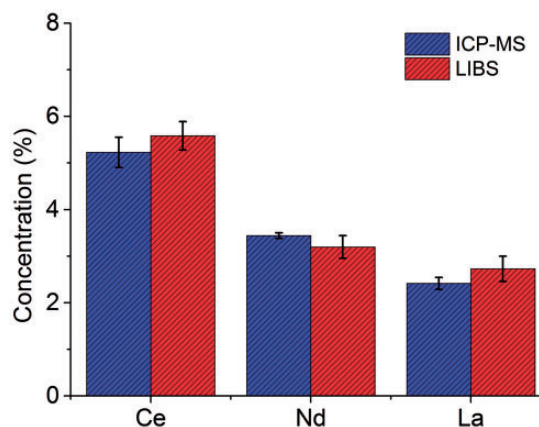


Figure 6. Graphical comparison of concentrations of Ce, La, and Nd calculated using ICP-MS and LIBS.

Conclusion

Different emission lines of REEs were identified in the spectra obtained from natural geological samples, which

confirmed the presence of those elements in corresponding samples. Ce, La, and Nd were detected in all the samples. Pr, Sm, Eu, Gd, and Dy were detected only in S8. Quantification of Ce, La, and Nd was performed by multivariate analysis. Partial least squares regression models were developed by using the Unscrambler software where seven samples were used to produce models and these models were used to predict the concentration of Ce, Nd, and La in a separate sample taken as an unknown. Prediction results were found to be comparable to those obtained by ICP-MS analysis. The study indicates that LIBS can be a potential technique for the detection and quantification of REEs in the geological samples.

Acknowledgments

The authors thank Drs. Harry M. Edenborn and Tracy Bank for their assistance with the ICP-MS analysis.

Conflict of Interest

The authors report there are no conflicts of interest.

Funding

This research was supported in part by an appointment of Mr. Bhatt to the National Energy Technology Laboratory Research Participation Program, sponsored by the U.S. Department of Energy, and administered by the Oak Ridge Institute for Science and Education.

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